

HAFSA FATHIMA

Artificial Intelligence & Machine Learning Undergraduate

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Education

Jayaprakash Narayana College of Engineering, Hyderabad

Expected 2027

Bachelor of Technology (B.Tech) in Artificial Intelligence & Machine Learning

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Machine Learning, Natural Language Processing, Deep Learning

Technical Skills

- **Programming Languages:** Python, Java, JavaScript, SQL, C
- **AI / Machine Learning:** PyTorch, TensorFlow, Scikit-learn, XGBoost, OpenCV, MediaPipe
- **LLM & AI Systems:** LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Multi-Agent Systems, Semantic Retrieval
- **Data Science & Analytics:** Pandas, NumPy, Matplotlib, Feature Engineering, Power BI
- **Backend & Deployment:** FastAPI, Node.js, REST APIs, Docker, Kafka, MLflow
- **Tools & Platforms:** Git, GitHub, Linux CLI, VS Code, Google Colab, PyCharm

Selected Projects

- **ReflexAI-Advanced – Multi-Agent AI Code Generation and Review Platform** GitHub
Multi-agent software engineering platform that transforms natural-language requirements into generated, reviewed, tested, secured, and documented code.
 - Designed a **6-agent workflow** using LangGraph and Groq LLMs for code generation, review, testing, performance analysis, security auditing, and documentation.
 - Implemented structured agent workflows to coordinate tasks across multiple stages of the software development lifecycle.
 - Integrated GitHub Pull Request automation and automated quality checks including code complexity analysis, test coverage assessment, and vulnerability detection.
- **Sentinel AI – Explainable Multi-Stage Fraud Detection System** GitHub
Explainable AI system for healthcare insurance claim fraud detection combining supervised learning, anomaly detection, deterministic rules, and an agentic review workflow.
 - Built an ensemble fraud-scoring pipeline using XGBoost, Isolation Forest, and deterministic fraud signals across **558K+ claims**, with provider-disjoint train/test splitting to reduce data leakage.
 - Achieved **81.3% precision, 75.4% recall, 78.2% F1-score, and 91.1% ROC-AUC** on the held-out test set, with an average scoring latency of **311 ms per claim**.
 - Developed a LangGraph workflow for claim preprocessing, risk scoring, validation, and LLM-generated explanations using Ollama, providing structured feature contributions for more auditable decisions.
- **AskDuo – Retrieval-Augmented AI Knowledge Assistant** GitHub
RAG-powered knowledge assistant that retrieves relevant information from internal documentation and generates source-grounded responses.
 - Built a modular Retrieval-Augmented Generation pipeline using LangChain, ChromaDB, OpenAI Embeddings, and GPT-4o-mini for semantic search across organizational knowledge bases.
 - Implemented document processing, persistent vector storage, semantic retrieval, and source-grounded response generation with citations for improved answer traceability.
- **Adaptive Streamer – Real-Time AI Anomaly Detection Platform** GitHub
Real-time anomaly detection system designed to identify abnormal patterns in streaming data and adapt to changing data distributions.
 - Built anomaly detection pipelines using CNN, LSTM, and TCN-based models, integrating ADWIN and Page-Hinkley methods for data drift detection.
 - Developed FastAPI APIs and a Streamlit dashboard for serving and visualizing predictions, with Kafka, Docker, and MLflow supporting streaming workflows, deployment, and experiment tracking.

Achievements

- Co-authored and published **2 machine learning research papers** in academic journals.
- Solved **100+ Data Structures and Algorithms problems on LeetCode**, covering arrays, strings, linked lists, trees, graphs, recursion, and dynamic programming.
- Presented AI/ML projects at **Science Fest 2025** and the **ESCI Hyderabad Tech Expo**.
- Developed and showcased AI-based solutions in multiple hackathons and technical competitions.

Research & Publications

- “*Car Price Prediction Using Machine Learning*”, published in *Research and Reviews: Advancement in Cyber Security* (2025).
- “*Future Proofing Real Estate: Machine Learning for Price Predictions*”, published in *Advancement of Computer Technology and Its Applications* (2025).

Hackathons & Competitions

- Participated in the **IARE Hackathon**, where I built **Sentinel AI**, an explainable multi-stage AI system for healthcare insurance claim fraud detection.
- Built **LegalShe**, an AI-powered multilingual legal assistance platform, during the **DevQueens Women’s Hackathon**.
- Participated in the **CBIT Hackathon** and **Specathon 2025**, developing AI-based solutions in time-constrained team environments.

Leadership & Activities

- **Technical Event Host & Anchor**: Hosted and moderated college-level technical events, managing speaker transitions, event flow, audience engagement, and coordination with organizing teams.
- **Hackathon Organizer**: Helped organize the college-level **Smart India Hackathon (SIH)** internal hackathon and another campus hackathon, coordinating registrations, participant support, event logistics, and execution.